

Evaluation of Offline-First Progressive Web Application Caching and Synchronization Strategies for Emergency Alerts in Low-Connectivity Environments

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Abstract— Communities in low-connectivity settings face major challenges in emergency notifications and resource access, posing a risk to their safety during critical situations. Previous approaches, such as SMS-based alert systems, lightweight native mobile apps, and centralized web platforms, have attempted to address these issues but remain dependent on continuous connectivity and fail under extended network outages. This study assessed the performance characteristics of two different Progressive Web App settings, namely Cache First, using instant synchronization, and Stale-While-Revalidate (SWR) using background internet synchronization. Network simulation settings using internet throttle were used for comprehensive browser testing, ensuring that both settings were analyzed using numerous tests. The major performance factors, cache hit ratio, synchronization latency, and offline submission success rates, were used for analysis. ANOVA and Bonferroni-adjusted pairwise comparisons were used for statistical analysis, and both tests clearly differentiated the two settings. Results obtained highlighted that Cache-First using instant synchronization possessed superior benefits for offline functionality, while SWR using background internet synchronization had quicker recovery during partial connectivity settings. These findings offer valuable direction for developers, government, and social organizations trying to develop emergency communication software for communities in low-connectivity settings.

Keywords— Background Synchronization, Cache-First, Emergency Alerts, Instant Synchronization, Low Connectivity, Offline-First, Progressive Web Applications, Stale-While-Revalidate

I. INTRODUCTION

Critical emergency communication for communities in peri-urban and rural settings around the world has remained inaccessible and unavailable mainly because of internet connectivity unreliability around the world [1, 2]. Networks, when untrustworthy and intermittent, can cause a time barrier in sending critical alerts and also restrictions in accessing critical resources for communities around the world [3, 4]. Some methods previously used for the same purpose, namely SMS alert systems, brief native mobile apps, and web applications, aimed at addressing the aforementioned issue, yet remained fully reliant on continuous internet connectivity for functionality, making it unsuitable for internet use during protracted internet outage situations [5-7]. While these solutions can provide basic communication functionality, they often fail under extended

network outages and offer limited support for offline data storage or synchronization once connectivity is restored.

Progressive Web Applications offer a viable solution by leveraging offline-first technologies, including service workers for background caching and resource loading, as well as persistent client-side storage through IndexedDB, which together enable resilient offline functionality [8-10]. For the purpose of caching, there are methods like network-first, cache-only, and SWR. Of the two, cache-first and SWR remain highly popular, making them extremely important in low-connectivity networks, especially when faster access is needed for both offline and real-time freshness [11]. Synchronization techniques for background processing are equally important, as they ensure reliable queueing and submission of data once connectivity is restored, preventing the loss of time-sensitive actions such as emergency alert submissions [12].

Despite prior research having largely focused upon related applications in the health and ecommerce industries, remarkably little has also discussed emergency communication systems upon communities [5, 7]. Emergency alerts, however, present distinct challenges: they are time-sensitive, require rapid and reliable delivery, and must remain functional under intermittent connectivity [6, 13, 14]. These considerations naturally led to the need for a clear understanding of the current limitations and gaps in emergency PWA implementations. The proposed study aims to address this gap by assessing both the performance outcomes for various types of caching and synchronizing algorithms for PWAs through simulations, facilitating an understanding upon their associated effectiveness for emergency communication operations upon communities for low-connectivity networks.

While PWAs offer offline-first capabilities through service workers and persistent storage, existing implementations rarely evaluate which combination of caching and synchronization strategies performs best under realistic low-connectivity scenarios [8-10]. PWAs have promise for effective emergency communication, but current deployments have challenges in delivering dynamic contents and synchronization within low-connectivity conditions [11, 12]. Cache-first strategies can improve offline availability but risk serving outdated information, whereas stale-while-revalidate (SWR) strategies maintain

fresher content yet depend on background fetches that may fail under unstable networks [15, 16]. Similarly, instant synchronization is beneficial in terms of reducing latency, but it is prone to failure in connectivity disruptions, while background synchronization with retries is beneficial in terms of reliability with increased latency [12, 17]. These limitations highlight a critical gap: empirical comparisons of caching and synchronization strategies, such as cache-first versus SWR paired with instant versus background synchronization, remain scarce under realistic emergency conditions. Scenarios involving sudden network outages, slow 2G/3G connections, queued multi-alert submissions, and concurrent resource retrieval have not been systematically evaluated [13, 18, 19]. Consequently, developers and policymakers lack evidence-based guidance on which architecture ensures timely, durable, and fresh alert delivery in low-connectivity communities [4, 14]. However, this study aims to evaluate and compare offline-first Progressive Web Application (PWA) caching and synchronization strategies to determine their effectiveness in supporting reliable delivery and submission of emergency alerts in low-connectivity environments.

II. LITERATURE REVIEW

Progressive Web Applications (PWAs) have received increasing attention for delivering near-native app experiences through web browsers, while supporting offline-first technologies such as service workers and IndexedDB [15, 20]. This offline-first design ensures that users can access content and perform actions even without connectivity, with updates synchronized once the connection is restored, a capability made possible through service workers that implement different caching strategies [9, 10]. For example, cache-first strategies serve content primarily from cache to reduce network use but may deliver outdated content [15], whereas stale-while-revalidate (SWR) strategies enhance user experience by delivering cached content instantly while updating it in the background to maintain data freshness [16]. Despite these advantages, comparative evaluations of these strategies under constrained connectivity remain limited [5, 8].

This limitation is particularly significant in the context of emergency communication, where timely and reliable alerts are critical. This is a major concern in emergency alert systems, as prompt and trustworthy alerts are crucial in such a situation. Conventional systems such as SMS or cellular notifications have been proved ineffective because of network congestion, infrastructure damage, or a lack of connectivity in rural areas of a country or region [6, 13]. Research also highlights disparities in connectivity between urban and rural communities, which further limits the effectiveness of digital warning systems [7, 18]. Existing mobile disaster applications frequently lack offline capabilities, relying heavily on uninterrupted connectivity for both alert dissemination and resource access [7, 18]. Together, these challenges underscore the need for resilient communication systems that can function reliably despite intermittent connectivity.

Despite growing interest in PWAs for emergency response, comparative studies of PWA architectures under constrained connectivity are scarce. While prior studies

have evaluated features such as GPS, push notifications, offline forms, or general performance in low-connectivity environments [8, 12, 18], none directly compare different service worker caching strategies or synchronization approaches under emergency conditions. For instance, Joseph, et al. [12] examine PWA performance in low-connectivity areas but do not test cache-first versus SWR strategies. Similarly, Syukron, et al. [18] and Cherukuri [8] focus on feature availability rather than architecture-level metrics such as cache hit ratio, synchronization latency, or offline submission success rates. This gap indicates that simulation-based evaluations are necessary to identify the most effective PWA configurations for emergency alerts and resource access in underserved communities.

Moreover, disaster risk management studies highlight the benefits of using design science approaches in designing and testing technological artifacts in a real-world setting [21, 22]. Integrating PWA design evaluation with performance analysis would thus represent a desirable strategy for improving disaster resilience framework approaches as well as helping in formulating an evidence-based emergency communications strategy [23, 24].

III. METHODOLOGY

A Design Science Research (DSR) paradigm and a simulation-based comparative methodology were used to evaluate different Progressive Web Application (PWA) models such as cache-first with instant sync and stale-while-revalidate with background sync for emergency alerts and resource access in low-connectivity communities. Synthetic PWAs were implemented in Vanilla JS to avoid framework-specific biases [25] and deployed on a local Node.js server with HTTPS. Testing was conducted on an Android emulator configured to match mid-range device specifications (Snapdragon 662, 4GB RAM, Chrome v125, Android 13) to reflect typical hardware in underserved areas [4, 26]. Network conditions were simulated using Chrome DevTools to replicate offline, 2G, and 3G throttling [3, 27].

Controlled browser-based simulations were then performed using network throttling to replicate offline and low-bandwidth (2G/3G) scenarios commonly experienced in underserved communities. Scenario design was derived from literature on low-connectivity emergency communication and consisted of repeated executions (minimum 30 runs per configuration) to ensure statistical reliability and reduce variability caused by network fluctuations or emulator performance [12, 18]. The three key performance metrics analyzed in this study were cache hit ratio, synchronization latency, and offline submission success rate.

The data was analyzed using ANOVA and t-test was used to assess caching strategies, cache-first and stale-while-revalidate (SWR), combined with synchronization methods, instant sync and background sync [11, 12]. Data from PWA logs were exported to IBM SPSS Statistics for repeated measures ANOVA to identify significant differences between strategies to identify the most effective method for community emergency applications [28, 29]. For factors showing significance, Bonferroni-adjusted pairwise comparisons were used to determine specific differences

between caching and synchronization strategies. While simulations provide accurate and reproducible outcomes, future field trials are recommended to validate results under real-world community conditions [19, 30].

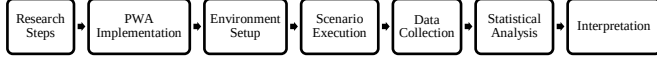


Fig. 1. Flow Diagram of Methodology

IV. RESULTS

A. Simulation Results

This section discusses the results obtained from evaluating both strategies using simulations, where it covered a comparison for Cache-First with Instant synchronization strategies, along with Stale-While- Revalidate (SWR) with Background Synchronization strategies for the three networks 2G, 3G, and Offline networks. The evaluation for the analysis includes important factors, namely cache hit ratio, synchronization latency, and offline submission success rate for the analysis. The factors used for evaluation are averages for 30 simulations for each configuration, using standard error (SE) to demonstrate variability for results.

The Cache Hit Ratio quantifies the amount of content served from the cache for analysis. Table 1 and Fig.2 showed Cache-First with Instant sync elicited higher cache hit ratios than SWR with Background Sync in offline network scenarios, while both strategies performed equally in 2G and 3G networks.

Table I. Cache Hit Ratio by Strategy and Network Condition

Network	Strategy	
	Cache-First + Instant Sync (Mean $\pm$ SE ^a)	SWR + Background Sync (Mean $\pm$ SE ^a)
2G	0.867 $\pm$ 0.033	0.867 $\pm$ 0.033
3G	0.867 $\pm$ 0.033	0.867 $\pm$ 0.033
Offline	0.867 $\pm$ 0.033	0.454 $\pm$ 0.034

^a SE : standard error of 30 simulation runs. Cache hit ratio measured as the proportion of requests served directly from cache during each simulation.

Cache Hit Ratio by Strategy and Network Condition

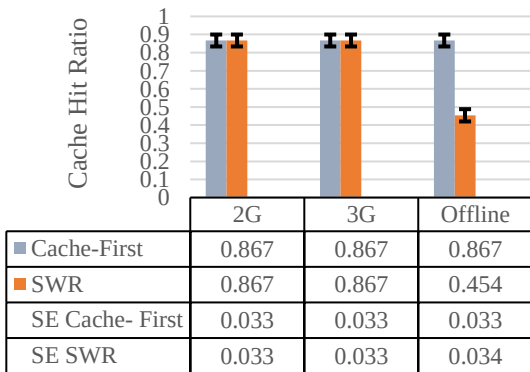


Fig. 2. Cache Hit Ratio by Strategy and Network Condition

Next, synchronization latency was evaluated to assess how quickly queued submissions were processed upon reconnection (Table 2, Fig. 3). Synchronization Latency is the average time taken by the aggregated submissions for synchronization upon reconnection. SWR with Background Sync had lower synchronization latency in both 2G and 3G networks, and Cache-First with Instant sync had similar latency during offline scenarios, having equal chances of instantaneous unsuccessful attempts during unstable networks.

Table II. Synchronization Latency by Strategy and Network Condition (ms)

Network	Strategy	
	Cache-First + Instant sync (Mean $\pm$ SE ^a)	SWR + Background Sync (Mean $\pm$ SE ^a)
2G	0 $\pm$ 11.709	364 $\pm$ 11.709
3G	0 $\pm$ 11.709	0 $\pm$ 11.709
Offline	0 $\pm$ 11.709	729 $\pm$ 11.91

^a SE : standard error of 30 simulation runs. Latency measured as mean transmission delay for successfully synchronized submissions.

Synchronization Latency by Strategy and Network Condition

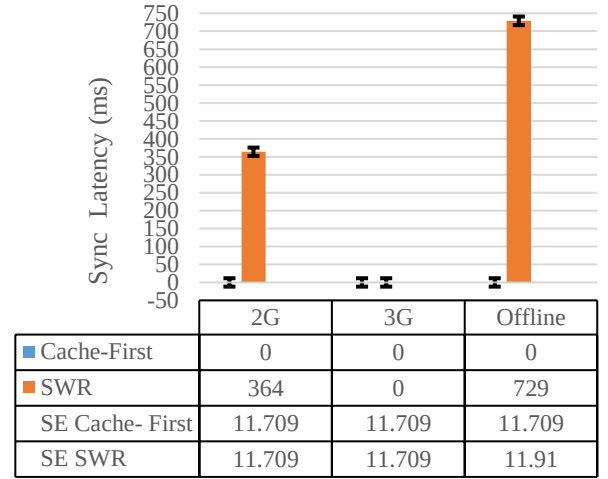


Fig. 3. Synchronization Latency by Strategy and Network Condition

Finally, the offline submission success rate was analyzed to determine which strategy maximized successful delivery during connectivity interruptions (Table 3, Fig. 4). Offline Submission Success Rate estimates the success rate of aggregated submissions for successful transmissions upon connection recovery. Both strategies were equal for 3G but Cache-First with Instant sync had better success rates in offline networks, while SWR with Background Sync had higher success rates during 2G networks, where its background validation feature impacts its success rate.

Table III. Offline Submission Success Rate by Strategy and Network Condition

Network	Strategy	
	Cache-First + Instant sync (Mean $\pm$ SE ^a)	SWR + Background Sync (Mean $\pm$ SE ^a)
2G	0 $\pm$ 0.909	1.43 $\pm$ 0.909
3G	0 $\pm$ 0.909	0 $\pm$ 0.909
Offline	15.5 $\pm$ 0.909	15 $\pm$ 0.925

^a SE : standard error of 30 simulation runs. Success rate measured as mean number of successful queued transmissions per simulation.

Offline Submissions by Strategy and Network Condition

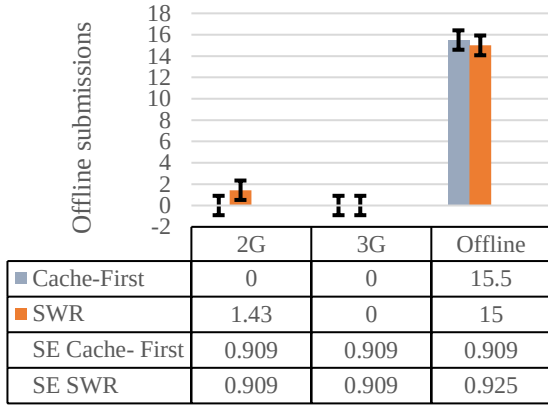


Fig. 4. Offline Submission Success Rate by Strategy and Network Condition

The Simulation Results clearly indicate that Cache-First with Instant Synchronization outperformed SWR with Background Synchronization in achieving higher cache hit ratios in offline networks. SWR had higher latency during partial connectivity (2G) networks and offline, while Cache-First had equal latency in all networks. The Offline submission success rates were higher for Cache-First in offline networks but lower with partial connectivity (2G). Initially, it gives an insight into how both methods will perform without any statistical analysis.

B. Statistical Analysis Results

Descriptive and inferential statistics were analyzed using IBM SPSS Statistics v29. ANOVA analysis was conducted to evaluate the effects of interaction, caching strategy, synchronization strategy, and network condition (2G, 3G, and Offline). Tests of normality and sphericity were satisfied (p -value > 0.05), confirming that the data met the assumptions for repeated measures analysis. For factors found to be significant, Bonferroni-adjusted pairwise comparisons were applied. The effect size, η^2 (eta squared), was also computed to indicate the strength of each effect. In the following analysis, F denotes the F -statistic from the ANOVA test, p -value represents the probability of observing the result by chance, and η^2 indicates the proportion of variance explained by each factor.

Cache Hit Ratio

In Table IV, analysis resulted in significant factors, Network Condition ($F(2,173)=25.798$, p -value < 0.05 , $\eta^2 = 0.230$) and Strategy ($F(1,173)=25.946$, p -value < 0.05 , $\eta^2 = 0.130$) for cache hit ratios, proving that both factors cause Cache-First and SWR + Instant sync to perform significantly better. The interaction factor was also found to significantly contribute ($F(2,173)=25.798$, p -value < 0.05) for cache hit ratios, proving that both strategies perform better in different networks. Pairwise Bonferroni tests discovered that there are significant differences from Offline to internet-connected networks, but not from 2G to 3G networks.

Table IV. ANOVA Results for Cache Hit Ratio

Source	ANOVA			
	F	p-value	η^2	Significance
Strategy	25.946	<0.05	0.130	**
Network	25.798	<0.05	0.230	**
Interaction	25.798	<0.05	—	**

** p -value < 0.05 , ns: not significant at p -value >0.05 .

Synchronization Latency

For synchronization latency, Table V shows there were significant main effects for both strategy ($F(1,173)=1445.068$, p -value < 0.05 , $\eta^2 = 0.893$) and network condition ($F(2,173)=480.960$, p -value < 0.05 , $\eta^2 = 0.848$). The interaction effect was also significant ($F(2,173)=480.960$, p -value < 0.05). Bonferroni pairwise comparison tests supported that SWR with Background Sync performed significantly faster for partial connectivity, although Cache-First had lower variability for both conditions.

Table V. ANOVA Results for Synchronization Latency

Source	ANOVA			
	F	p-value	η^2	Significance
Strategy	1445.068	<0.05	0.893	**
Network	480.960	<0.05	0.848	**
Interaction	480.960	<0.05	—	**

** p -value < 0.05 , ns: not significant at p -value >0.05 .

Offline Submission Success Rate

The analysis showed that there was a significant effect for network condition ($F(2,173)=177.099$, p -value < 0.05 , $\eta^2 = 0.672$) but no significant main effect for strategy ($F(1,173)=0.175$, p -value $= 0.677$) and interaction ($F(2,173)=177.099$, p -value > 0.05) (Table VI). The Bonferroni tests showed that the offline success rates for both strategies were significantly higher for Offline than for 2G and 3G networks, while no significant difference was found for both strategies.

Table VI. ANOVA Results for Offline Submission Success Rate

Source	ANOVA			
	F	p-value	η^2	Significance
Strategy	0.175	0.677	—	ns
Network	177.099	< 0.05	0.672	**
Interaction	177.099	> 0.05	—	ns

** p -value < 0.05 , ns: not significant at p -value >0.05 .

Summary of Findings

The results indicate that both caching strategy and network condition had a significant effect on cache hit ratio and synchronization latency as determined by the ANOVA test results (Table IV & Table V, Fig. 2 & Fig. 3). Cache-First with Instant Sync had the best results in terms of reliability during offline activities as indicated by the results (Table I & Table VI, Fig. 2 & Fig. 4). SWR with Background Sync had slightly improved synchronization latency during 3G network connectivity (Tables II & Table V, Fig. 3) and improved submission success rate during 2G networks (see Table III & Table 6, Fig. 4).

V. DISCUSSION

This study analyzed the usage of the first caching strategy, Cache First with Instant sync, as well as SWR strategy, SWR with Background sync, in Progressive Web Applications when it comes to low connectivity networks (2G networks, 3G networks, and Offline networks) through simulation experiments.

The findings illustrated that Cache-First with Instant Sync presented greater cache hit ratio and reliability during the offline process, particularly for fully offline networks, proving its robustness for use during low-connectivity scenarios. This outcome directly aligns with the study's objectives, confirming that Cache-First can sustain accessibility during complete disconnection. It also reinforces related studies that signal that first caches, mainly SWR, can conduct enhanced performance during low-connectivity networks when background synchronization is supported [10, 12, 15].

However, the SWR strategy during fully disconnected networks illustrated lower reliability for the offline process, proving its need for continuous connectivity during actual internet scenarios. This highlights a distinct trade-off between reliability and synchronization efficiency depending on network conditions. The results further reinforce related studies that signal its dependency on constant internet connectivity for enhanced performance [8, 16]. The interaction effects shown by ANOVA also support that the effectiveness of the caching method is highly dependent on the network, confirming that performance outcomes vary according to connection stability.

The results obtained from the experiment indicate that Cache-First with Instant Sync prioritizes reliability and availability for accessing data, and it is therefore highly recommended for emergency communication apps where continuous functionality is essential. This reinforces the method's suitability for environments where uninterrupted access to alerts is critical. Conversely, the SWR with Background Sync exhibited lower latency under partial networks and can therefore benefit user experience, although it may compromise reliability under unstable connections. This demonstrates that both methods serve distinct purposes depending on the intended application context.

For developers and policymakers, Cache-First with Instant Sync can act as a practical recommendation for the use of PWAs in low-connectivity regions during emergencies, ensuring that alerts reach citizens every time. This provides clear guidelines for technical implementation and deployment in resource-constrained communities. On the other hand, SWR can perform better in urban or semi-connected areas, where connectivity is intermittent yet generally consistent. These insights emphasize the contextual adaptability of caching strategies in real-world emergency communication systems.

Although it was possible to conduct a controlled experiment using simulations for evaluating caching methods and synchronization strategies, it has certain limitations. The study showed that simulations carried out using Chrome DevTools with an Android emulator, had limited factors such as the number of networks and without simulations

related to other factors that could occur in reality, like internet heterogeneity, performance for different gadgets, and shared usage by numerous people. Only the three factors, being cache hit ratio, synchronization latency, and offline submission success rate, were analysed, while other aspects such as energy efficiency and user experience were beyond the study's scope. Moreover, simulations were not carried out for random factors like simultaneous emergency alert submissions, user mobility, and partially connected regions. These limitations suggest areas for refinement and extension in future work, as the findings, though indicative, reflect controlled simulations rather than real-world implementations.

VI. CONCLUSIONS

This study aimed to evaluate low-connectivity Progressive Web Apps for emergency alerts using simulation-based testing of caching strategies under 2G, 3G, and Offline networks. The study focused on assessing the performance of Cache-First with Instant Synchronization and SWR with Background Synchronization based on three key metrics: cache hit ratio, synchronization latency, and offline submission success rate. By systematically analysing these metrics, the experiment provided insights into how caching and synchronization strategies influence reliability and responsiveness under varying connectivity conditions.

The results indicated that Cache-First strategy produced superior reliability compared to SWR, particularly in offline environments, while maintaining a stable cache hit ratio across all networks. These findings confirm Cache-First's suitability for continuous alert delivery during network outages. In contrast, SWR demonstrated slightly faster synchronization in partially connected conditions (2G and 3G) but struggled with lower offline submission success rates due to its dependency on live background revalidation. This reinforces the observed interaction effects, which indicate that caching performance is highly sensitive to network availability and should be optimized according to connectivity context.

For real world implementation, developers, NGOs, and policymakers can use this method for designing emergency PWAs for low-connectivity communities. Cache-First combined with instant synchronization has shown to maximize reliability and data retention when internet access is unavailable; this approach can enhance the dependability of emergency alert systems, ensuring consistent delivery of critical information even during disconnections.

Future research should extend these findings through real-world field evaluations, incorporating additional parameters such as device diversity, concurrent user behaviour, and adaptive caching or notification mechanisms. By integrating these elements, future systems can further strengthen the resilience and responsiveness of emergency PWAs, supporting communities in low-connectivity regions through timely alerts and reliable resource access.

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